

ABSTRACT ALGEBRA 1 - HOMEWORK 3- M. VICARÍA

Dear student please submit the complete solution to each of the following problems in Gradescope in a PDF file. Please write down your name in each page and specify clearly the problem that you stated. For your study purposes, I included a classification of the problems and their difficulties:

- Warm-up: these are require that you understand some definition.
- *: they indicate some medium level of difficulty. You should spend longer in these problems.
- ** indicates that the problem is challenging, they will require more time than the other problems.

The complexity from ** and * is the level we would like you to master in this course. Questions highlighted with **OPTIONAL** are not necessary to get full credit for the homework, but I highly encourage you to solve them for your learning process.

In each of the solutions you **SHOULD JUSTIFY COMPLETELY YOUR ANSWER** with a proof or a counter-example. No credit is granted to solutions without justification.

1. MANDATORY PROBLEMS

- (1) **Problem 1 (1.5 points):** Let $(G, \cdot)$ be a group.
- (a) (1 points) Show that $(G, \cdot)$ is abelian if and only if the map $g \mapsto g \cdot g$ is a group homomorphism.
 - (b) (0.5 points) Assume that for all $x \in G$, $x \cdot x = 1$. Show that G is abelian.

(2) **Problem 2** : (6.5 points) Groups and automorphisms** Let $(G, \cdot, e)$ be a group and

$$\text{Aut}(G) := \{f : G \rightarrow G \mid f \text{ is an isomorphism}\},$$

and let $\circ : \text{Aut}(G) \times \text{Aut}(G) \rightarrow \text{Aut}(G)$ be the binary operation defined by sending a pair (f, g) to $f \circ g$.

- (a) (1.5 point) Show that $(\text{Aut}(G), \circ)$ is a group.
- (b) (1 points) For any $g \in G$, we can define a function $\phi_g : G \rightarrow G$ given by $\phi_g(h) = ghg^{-1}$. Show that ϕ_g is a bijective group homomorphism (i.e., an *automorphism*) for any $g \in G$. Any automorphism of this form is called an *inner* automorphism.
- (c) (2 points) Let $\phi : G \rightarrow \text{Aut}(G)$ be the map given by $\phi(g) = \phi_g$. Show that ϕ is a group homomorphism, where the operation on the codomain is function composition.
- (d) (1 point) Let $\text{End}(G) = \{\phi_g \in \text{Aut}(G) \mid g \in G\}$ i. e. the set of inner automorphisms. Show that $\text{End}(G)$ is a subgroup of $(\text{Aut}(G), \circ)$.
- (e) (1 points) Assume $\text{End}(G)$ is trivial, i.e. it is $\{Id_G\}$. Show that G must be abelian.

- (3) **Problem 3*:** (2 points) Let $(G, \cdot)$ be a group and $H \subseteq G$ a subset. Let $\equiv_H$ be the relation on G given by

$$a \equiv_H b \iff ab^{-1} \in H.$$

Show that if $\equiv_H$ is an equivalence relation on G , then $H \leq G$ is a subgroup.

2. OPTIONAL PROBLEMS

- (1) **Optional Problem 1*:** We consider $G = \text{GL}_2(\mathbb{R})$ together with matrix multiplication. Check whether the following

subsets are subgroups and, if they are, that they are abelian:

- $H_1 = \{A \in G \mid A \text{ diagonal}\} \subseteq G$,
- $H_2 = \left\{ \begin{pmatrix} a & b \\ c & 1 \end{pmatrix} \mid a, b, c \in \mathbb{R}, a - bc \neq 0 \right\} \subseteq G$.

Can you give an (other) example of a proper, non-abelian subgroup?

- (2) **Optional Problem 2**:**

Let $Y \subset X$ be sets, let $(G, *)$ be a group and let G^X be the set of functions $X \rightarrow G$. We define an operation $*$ on G^X pointwise, i.e. for $f: X \rightarrow G$ and $g: X \rightarrow G$, the function $f * g: X \rightarrow G$ is defined via $(f * g)(x) := f(x) * g(x)$. Let

$$N := \{f \in G^X : f(y) = 1 \text{ for all } y \in Y\} \subseteq G^X,$$

where 1 is the neutral element in $(G, *)$.

- (a) (3 point) Show that $(G^X, *)$ is a group. What is the neutral element?
 (b) (2 point) Show that

$$\begin{aligned} \phi: G^X &\rightarrow G^Y, \\ f &\mapsto f|_Y \end{aligned}$$

is a group homomorphism.